

```
[1]: import pandas as pd

df = pd.read_csv('googleplaystore.csv')

df.head()
```

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0.0	4.0.3 and up
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0.0	4.0.3 and up
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Everyone	Art & Design	August 1, 2018	1.2.4	4.0.3 and up
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	Teen	Art & Design	June 8, 2018	Varies with device	4.2 and up
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Everyone	Art & Design;Creativity	June 20, 2018	1.1	4.4 and up

```
[2]: df[df.duplicated(subset='App', keep=False)]
```

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0.0	4.0.3 and up
23	Mcqueen Coloring pages	ART_AND_DESIGN	NaN	61	7.0M	100,000+	Free	0	Everyone	Art & Design;Action & Adventure	March 7, 2018	1.0.0	4.1 and up
36	UNICORN - Color By Numbers	ART_AND_DESIGN	4.7	8145	24M	500,000+	Free	0	Everyone	Art & Design	August 2, 2018	1.0.9	4.4 and up

42	Textgram - write on photos	ART_AND_DESIGN	4.4	295221	Varies with device	10,000,000+	Free	0	Everyone	Art & Design	July 30, 2018	Varies with device	Varies with device
139	Wattpad  Free Books	BOOKS_AND_REFERENCE	4.6	2914724	Varies with device	100,000,000+	Free	0	Teen	Books & Reference	August 1, 2018	Varies with device	Varies with device
...	...	...	...	...	...	...	...	...	...	...	...	...	...
10715	FarmersOnly Dating	DATING	3.0	1145	1.4M	100,000+	Free	0	Mature 17+	Dating	February 25, 2016	2.2	4.0 and up
10720	Firefox Focus: The privacy browser	COMMUNICATION	4.4	36981	4.0M	1,000,000+	Free	0	Everyone	Communication	July 6, 2018	5.2	5.0 and up
10730	FP Notebook	MEDICAL	4.5	410	60M	50,000+	Free	0	Everyone	Medical	March 24, 2018	2.1.0.372	4.4 and up
10753	Slickdeals: Coupons & Shopping	SHOPPING	4.5	33599	12M	1,000,000+	Free	0	Everyone	Shopping	July 30, 2018	3.9	4.4 and up
10768	AAFP	MEDICAL	3.8	63	24M	10,000+	Free	0	Everyone	Medical	June 22, 2018	2.3.1	5.0 and up

1979 rows × 13 columns

```
[13]: #check duplicates
df["App"].duplicated().sum()
```

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[13]: np.int64(698)
```

```
[14]: df[df["App"].duplicated()]
```

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
285	Quick PDF Scanner + OCR FREE	BUSINESS	4.2	80804	Varies with device	5,000,000+	Free	0	Everyone	Business	February 26, 2018	Varies with device	4.0.3 and up

293	PDF Editor	BUSINESS	4.2	100000	5M	100,000,000+	Free	0	Everyone	Business	2018	Varies with device	up
294	Slack	BUSINESS	4.4	51510	Varies with device	5,000,000+	Free	0	Everyone	Business	August 2, 2018	Varies with device	Varies with device
382	Messenger – Text and Video Chat for Free	COMMUNICATION	4.0	56646578	Varies with device	1,000,000,000+	Free	0	Everyone	Communication	August 1, 2018	Varies with device	Varies with device
383	imo free video calls and chat	COMMUNICATION	4.3	4785988	11M	500,000,000+	Free	0	Everyone	Communication	June 8, 2018	9.8.000000010501	4.0 and up
...	...	...	...	...	...	...	...	...	...	...	...	...	...
10647	Motorola FM Radio	VIDEO_PLAYERS	3.9	54815	Varies with device	100,000,000+	Free	0	Everyone	Video Players & Editors	May 2, 2018	Varies with device	Varies with device
10715	FarmersOnly Dating	DATING	3.0	1145	1.4M	100,000+	Free	0	Mature 17+	Dating	February 25, 2016	2.2	4.0 and up
10720	Firefox Focus: The privacy browser	COMMUNICATION	4.4	36981	4.0M	1,000,000+	Free	0	Everyone	Communication	July 6, 2018	5.2	5.0 and up
10730	FP Notebook	MEDICAL	4.5	410	60M	50,000+	Free	0	Everyone	Medical	March 24, 2018	2.1.0.372	4.4 and up
10753	Slickdeals: Coupons & Shopping	SHOPPING	4.5	33599	12M	1,000,000+	Free	0	Everyone	Shopping	July 30, 2018	3.9	4.4 and up

698 rows × 13 columns

```
[15]: #View all occurrences of duplicate apps
df[df["App"].duplicated(keep=False)].sort_values("App")
```

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
5415	365Scores - Live Scores	SPORTS	4.6	666246	25M	10,000,000+	Free	0	Everyone	Sports	July 29, 2018	5.5.9	4.1 and up

698 rows × 13 columns

```
[15]: #View all occurrences of duplicate apps
df[df["App"].duplicated(keep=False)].sort_values("App")
```

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
5415	365Scores - Live Scores	SPORTS	4.6	666246	25M	10,000,000+	Free	0	Everyone	Sports	July 29, 2018	5.5.9	4.1 and up
3083	365Scores - Live Scores	SPORTS	4.6	666521	25M	10,000,000+	Free	0	Everyone	Sports	July 29, 2018	5.5.9	4.1 and up
1755	8 Ball Pool	GAME	4.5	14200344	52M	100,000,000+	Free	0	Everyone	Sports	July 31, 2018	4.0.0	4.0.3 and up
1703	8 Ball Pool	GAME	4.5	14198602	52M	100,000,000+	Free	0	Everyone	Sports	July 31, 2018	4.0.0	4.0.3 and up
1871	8 Ball Pool	GAME	4.5	14201891	52M	100,000,000+	Free	0	Everyone	Sports	July 31, 2018	4.0.0	4.0.3 and up
...	...	...	...	...	...	...	...	...	...	...	...	...	...
2590	textPlus: Free Text & Calls	SOCIAL	4.1	382120	28M	10,000,000+	Free	0	Everyone	Social	July 26, 2018	7.3.1	4.1 and up
2964	theScore: Live Sports Scores, News, Stats & Vi...	SPORTS	4.4	133825	34M	10,000,000+	Free	0	Everyone 10+	Sports	July 25, 2018	6.17.2	4.4 and up
3055	theScore: Live Sports Scores, News, Stats & Vi...	SPORTS	4.4	133833	34M	10,000,000+	Free	0	Everyone 10+	Sports	July 25, 2018	6.17.2	4.4 and up
3652	wetter.com - Weather and Radar	WEATHER	4.2	189313	38M	10,000,000+	Free	0	Everyone	Weather	August 6, 2018	Varies with device	Varies with device
8291	wetter.com - Weather and Radar	WEATHER	4.2	189310	38M	10,000,000+	Free	0	Everyone	Weather	August 6, 2018	Varies with device	Varies with device

1221 rows × 13 columns

```
[17]: #Remove duplicate apps
```

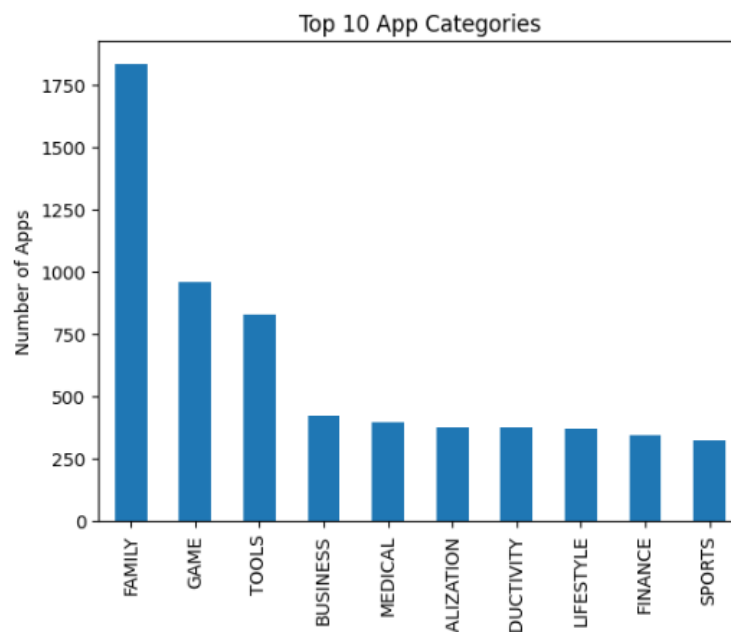
```
[18]: df["App"].duplicated().sum()
```

```
[18]: np.int64(0)
```

```
[19]: #Top 10 Categories Chart
import matplotlib.pyplot as plt

df["Category"].value_counts().head(10).plot(kind="bar")

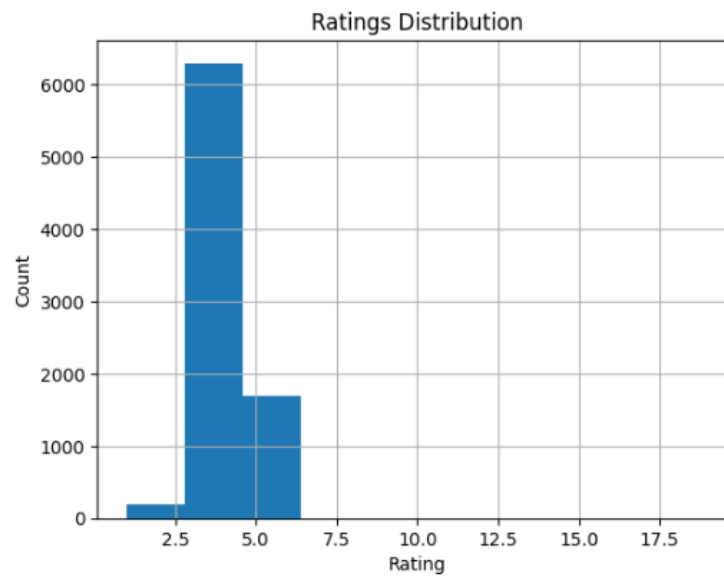
plt.title("Top 10 App Categories")
plt.xlabel("Category")
plt.ylabel("Number of Apps")
plt.show()
```



Category

```
[20]: #Ratings Distribution
df["Rating"].hist()

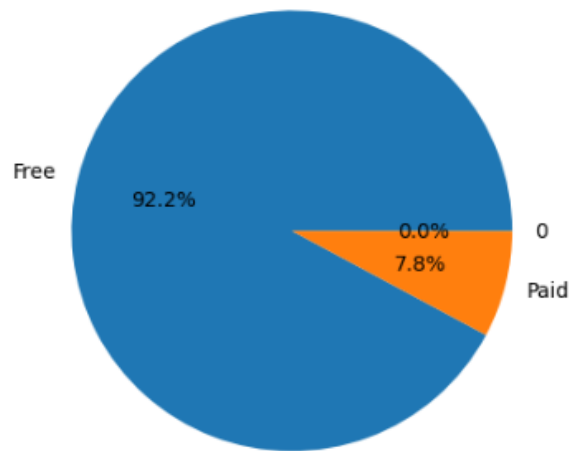
plt.title("Ratings Distribution")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.show()
```



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[24]: df["Type"].value_counts().plot(kind="pie", autopct="%1.1f%%")

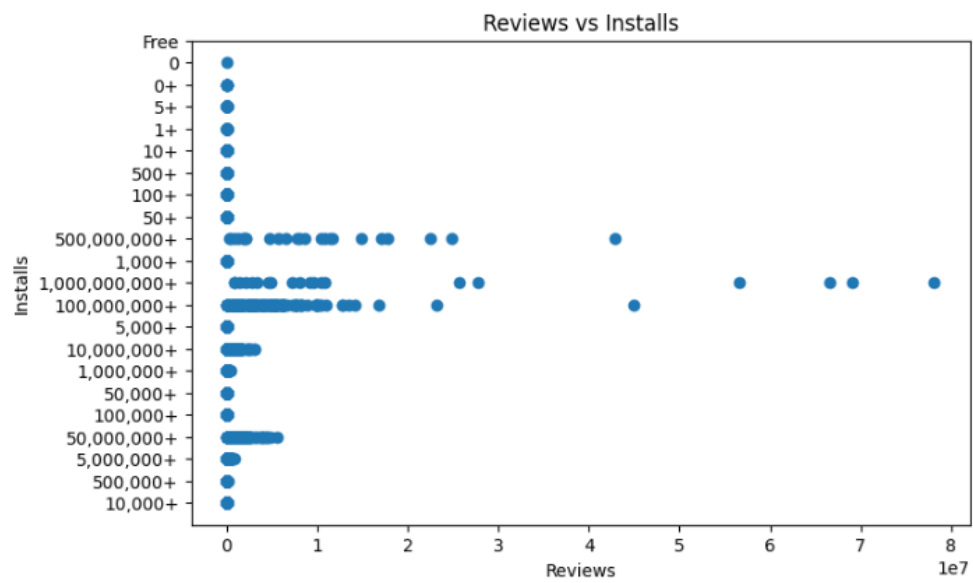
plt.title("Free vs Paid Apps")
```

Free vs Paid Apps



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[29]: plt.figure(figsize=(8,5))
plt.scatter(df['Reviews'], df['Installs'])

plt.title('Reviews vs Installs')
plt.xlabel('Reviews')
plt.ylabel('Installs')
plt.show()
```



```
[31]: #Label encoding
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df['Type_Encoded'] = le.fit_transform(df['Type'])

print(df[['Type', 'Type_Encoded']].head())
```

0	Free	1
1	Free	1
2	Free	1
3	Free	1
4	Free	1

```
[35]: from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df['Type_Encoded'] = le.fit_transform(df['Type'])
df['Content_Rating_Encoded'] = le.fit_transform(df['Content Rating'])
print(df.head())
```

	App	Category	Rating \
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1
1	Coloring book moana	ART_AND_DESIGN	3.9
2	U Launcher Lite - FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3

	Reviews	Size	Installs	Type	Price	Content	Rating \
0	159.0	19M	10,000+	Free	0	Everyone	
1	967.0	14M	500,000+	Free	0	Everyone	
2	87510.0	8.7M	5,000,000+	Free	0	Everyone	
3	215644.0	25M	50,000,000+	Free	0	Teen	
4	967.0	2.8M	100,000+	Free	0	Everyone	

	Genres	Last Updated	Current Ver \
0	Art & Design	January 7, 2018	1.0.0
1	Art & Design;Pretend Play	January 15, 2018	2.0.0
2	Art & Design	August 1, 2018	1.2.4
3	Art & Design	June 8, 2018	Varies with device
4	Art & Design;Creativity	June 20, 2018	1.1

	Android Ver	Type_Encoded	Content_Rating_Encoded
0	4.0.3 and up	1	1
1	4.0.3 and up	1	1
2	4.0.3 and up	1	1
3	4.2 and up	1	4
4	4.4 and up	1	1